Individual Honours Project

WEEKLY REPORT

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UIT - BCU

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I. Summary

- This is the report of my project "**BCG: A Game-Based Approach to Motor Imagery Adaptation and BCI Calibration.**"
- The target was to make EEG-based motor imagery calibration more effective and easier to apply outside the laboratory.

- Timeline:

STP7	**Dataset preparation and model training**	**18-Mar-26**	**29-Mar-26**	**12**	**8**	**Passed 3d**	**70%**
1	Identify and select appropriate public MI datasets				-		**100%**
2	Design EEG preprocessing and signal cleaning pipeline				-		**100%**
3	Exploratory data analysis				-		**100%**
4	Train and compare candidate classification models				-		**50%**
5	Select final model for proof-of-concept deployment				-		**0%**
STP8	**Game development**	**30-Mar-26**	**13-Apr-26**	**15**	**11**	**In progress**	**0%**
1	Develop game prototype				-		**0%**
2	Add calibration mechanics				-		**0%**
3	Design EEG-driven game controls				-		**0%**
4	Design game feedback and progression system				-		**0%**
5	Refine game usability and calibration user experience				-		**0%**

III. Current Difficulties

- The results of trained models are very low:

Model	Dataset	Evaluation	Accuracy
CSP + LDA	MIMED	Cross-subject, 5-fold	52.5%
CSP + SVM	MIMED	Cross-subject, 5-fold	46.7%
CSP + LDA	MIMED	Within-subject	42.1%

- Possible cause: Dataset too small overall (240 epochs total)
- Next we will head into possible solutions

III. Current Difficulties

Current dataset: MIMED, EMOTIV Epoc, 14-channel, 30 subjects, 240 trials total

Solution A:

- Try to train more models using a bigger dataset, then evaluate them
- PhysioNet EEGMMIDB (2009): 64-channel, 109 subjects, ~9800 trials total, widely used benchmark
- OpenBMI (2019): 62-channel, 54 subjects, 10800 trials total, very large (~200GB)

Solution B:

- Use a pretrained model EEGNetv4, trained on large-scale EEG benchmark data
- Fine-tune using data collected with EMOTIV Epoc X
- But I am not sure if this way is fine for the Honours Project or any publication

THANK YOU FOR READING!

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